Idea Proposal: 1–3 Day-Ahead Solar Power Forecasting for Türkiye

1. Project Idea

Türkiye's installed solar capacity is growing fast, and the grid operator and market participants must plan generation days in advance. Day-ahead forecasting is common, but a leakage-free, measured evaluation for 2–3 days ahead is missing.

Our goal is to build a model that forecasts the total hourly generation of Türkiye's licensed solar plants 24, 48 and 72 hours ahead with an 80% uncertainty interval, plus a demo anyone can try in the browser. We will measure success on daylight hours with nMAE, a skill score against simple reference methods, and the coverage rate of the interval.

2. Relevance to SDGs

SDG 7.2 (increase the share of renewable energy): More reliable multi-day forecasts help the grid absorb solar power with less reserve capacity and lower imbalance costs.
SDG 13 (climate action): Less need for fossil-fuelled backup generation gives an indirect emissions benefit.

3. Literature Examples

- **Ertekin (2020)**, Hittite J. Sci. Eng. 7(1), 35–40: Forecasts Türkiye's total solar generation (2017–2019) one day ahead with an ensemble that combines EMD decomposition with ANN, SVR and GBM; shows that hour-based modelling performs better. Gap: 24-hour horizon only, no uncertainty interval.
- **Biçer (2026)**, MSc thesis, Istanbul Technical University: Day-ahead forecasts for three plants in Konya, Sivas and Antalya using ECMWF IFS forecasts, XGBoost and quantile regression; best nMAE 5.2%. Reports that tree-based quantile models produce overly narrow intervals. Gap: plant level, day-ahead only.

Our contribution: national scale, 1–3 day horizons and a calibrated uncertainty interval.

4. Data

- **Target:** EPİAŞ Transparency Platform, hourly licensed solar generation (API, JSON → CSV/Parquet), converted to capacity factor by dividing by licensed installed capacity.
- **Inputs:** Open-Meteo Previous Runs API; hourly weather forecasts issued 1, 2 and 3 days earlier (irradiance, cloud cover, temperature) at 10–15 points in the provinces where licensed plants are concentrated.
- **Size:** January 2024 – October 2026, about 34 months and 24,000 hourly rows; no manual labelling needed.
- **Preprocessing:** Time-zone alignment, flagging missing and anomalous hours, derived features such as solar elevation and clear-sky index, excluding night hours from evaluation, time-based split (last 12 months as test) and scaling fitted on training data only.

5. Approach: Machine Learning

We chose machine learning because the data is tabular and medium-sized: gradient boosting models are strong on this structure, train quickly without a GPU and can be explained with SHAP, while deep learning would add cost without clear benefit at this data size. We will first build reference models (persistence, climatology and their combination, CLIPER); the main model is a separate **LightGBM** for each horizon. For the uncertainty interval we will use quantile LightGBM with conformal calibration. Demo: Streamlit on Hugging Face Spaces.